Quiz 4 (Version 1)

CAS CS 132: *Geometric Algorithms*

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- ▷ You will have approximately 35 minutes to complete this exam.
- ▷ Your final solution must appear in the solution boxes for each problem. **Only include your final solution in the solution box. You must show your work outside of the solution box.** You will not recieve credit it you don't show your work.

1 Matrix Inverses

$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 1 \\ -2 & 5 & -1 \end{bmatrix}$$

Determine the inverse of the above matrix.

$$\left[\begin{array}{cccccc} 1 & -1 & 1 & 1 & 0 & 0 \\ \overset{-1}{1} & \overset{+1}{0} & \overset{-1}{1} & \overset{-1}{0} & 1 & 0 \\ \overset{+2}{-2} & \overset{-2}{5} & \overset{+2}{-1} & \overset{+2}{0} & 0 & 1 \end{array}\right] \sim \left[\begin{array}{cccccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & \overset{-3}{3} & 1 & \overset{+3}{2} & \overset{-3}{0} & 1 \end{array}\right] \sim \left[\begin{array}{cccccc} 1 & -1 & \overset{-1}{1} & \overset{-5}{1} & \overset{+3}{0} & \overset{-1}{0} \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 5 & -3 & 1 \end{array}\right]$$

$$\sim \left[\begin{array}{cccccc} 1 & \overset{+1}{-1} & 0 & \overset{-1}{-4} & \overset{+1}{3} & -1 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 5 & -3 & 1 \end{array}\right] \sim \left[\begin{array}{cccccc} 1 & 0 & 0 & -5 & 4 & -1 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -5 & -3 & 1 \end{array}\right]$$

Solution.

$$\begin{bmatrix} -5 & 4 & -1 \\ -1 & 1 & 0 \\ -5 & -3 & 1 \end{bmatrix}$$

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2 Steady-State Vectors

$$T = \begin{bmatrix} 0.3 & 0.9 & 0 & 0 \\ 0.7 & 0.1 & 0 & 0 \\ 0 & 0 & 0.2 & 0.4 \\ 0 & 0 & 0.8 & 0.6 \end{bmatrix}$$

Determine a general form solution which describes the steady-state vectors of the above matrix T. Recall that $\mathbf{v}$ is a steady-state vector of T if $T\mathbf{v} = \mathbf{v}$.

$$T - I = \begin{bmatrix} -0.7 & 0.9 & 0 & 0 \\ 0.7 & -0.9 & 0 & 0 \\ 0 & 0 & -0.8 & 0.4 \\ 0 & 0 & 0.8 & -0.4 \end{bmatrix} \sim \begin{bmatrix} 1 & -9/7 & 0 & 0 \\ 0 & 0 & 1 & -1/2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution.

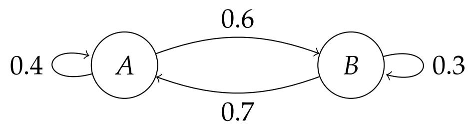
$x_1 = 9/7\, x_2$

x_2 is free

$x_3 = 1/2\, x_4$

x_4 is free

3 Steady-State Distribution



Determine the steady-state distribution for the Markov chain given by the above state diagram. Mark in your solution which component in of the distribution corresponds to which state in the diagram.

$$T = \begin{array}{c c} & \begin{array}{cc} A & B \end{array} \\ \begin{array}{c} A \\ B \end{array} & \begin{bmatrix} 0.4 & 0.7 \\ 0.6 & 0.3 \end{bmatrix} \end{array}$$

$$T - \underline{I} = \begin{bmatrix} -0.6 & 0.7 \\ 0.6 & -0.7 \end{bmatrix} \sim \begin{bmatrix} 1 & -7/6 \\ 0 & 0 \end{bmatrix}$$

$x_1 = 7/6\, x_2$

x_2 is free

$$x_1 + x_2 = 1$$

$$\frac{7}{6} x_2 + x_2 = 1$$

$$\frac{13}{6} x_2 = 1 \qquad x_2 = 6/13$$

Solution.

$$\begin{array}{c c} \begin{array}{c} A \\ B \end{array} & \begin{bmatrix} 7/13 \\ 6/13 \end{bmatrix} \end{array}$$